

PALU MUTIARA, Indonesia

WMO: 970720

Lat: 0.9159S

Lon: 119.9053E

Elev: 20

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	72.2	73.0	65.2	93.2	85.4	66.8	98.8	84.5	20.3	87.6	18.5	88.4	1.0	130	0.338

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	15.5	94.7	77.3	93.6	77.3	92.6	77.3	79.9	89.6	79.3	88.7	79.0	88.3	13.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	142.7	83.0	77.0	140.8	82.7	76.5	138.3	82.3	43.4	90.1	42.9	89.4	42.4	88.6	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.7	15.8	69.6	97.4	1.5	1.6	68.5	98.5	67.6	99.4	66.7	100.3	65.6	101.4		
(5)			66.1	82.2	2.1	1.8	64.6	83.5	63.4	84.6	62.2	85.6	60.7	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	81.9	82.1	82.7	82.9	83.1	82.1	81.8	82.2	82.8	83.1	82.7	82.3	(6)
(7)		DBStd	2.03	1.94	2.06	2.14	1.99	1.84	1.92	2.01	2.04	1.86	1.96	1.91	2.11	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11855	988	899	1013	988	1025	963	985	999	983	1027	982	1003	(10)
(11)		CDD65	6380	523	479	548	538	560	513	520	534	533	562	532	538	(11)
(12)		CDH74	64969	5159	4755	5626	5649	5887	5029	4966	5290	5514	6017	5548	5529	(12)
(13)		CDH80	25014	1819	1761	2204	2227	2332	1852	1823	2063	2250	2472	2142	2069	(13)
(14)	Wind	WSAvg	4.6	4.8	4.9	5.2	4.6	4.3	4.0	4.2	4.6	4.9	5.0	4.8	4.5	(14)
(15)	Precipitation	PrecAvg	47.8	4.2	3.3	3.8	4.5	4.9	4.7	5.0	3.7	2.8	3.6	3.8	4.1	(15)
(16)		PrecMax	66.9	9.6	6.4	8.6	11.0	8.3	10.0	11.1	11.7	5.4	9.8	7.0	11.3	(16)
(17)		PrecMin	32.6	2.3	1.7	1.8	2.0	2.5	1.1	1.6	0.5	0.2	1.0	2.0	1.3	(17)
(18)		PrecStd	8.3	1.9	1.2	1.7	1.7	1.5	1.7	2.4	2.7	1.4	2.0	1.2	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.1	94.4	95.1	94.9	94.9	94.0	93.6	94.4	94.9	95.4	94.3	94.4	(19)
(20)			MCWB	77.9	76.9	76.9	77.6	78.3	77.3	77.4	77.0	76.9	77.1	77.4	77.6	(20)
(21)		2%	DB	91.4	92.3	93.1	92.9	92.7	92.0	91.6	92.6	93.1	93.6	92.6	92.2	(21)
(22)			MCWB	77.0	77.0	77.0	77.6	78.2	77.3	77.2	76.7	76.8	77.1	77.4	77.3	(22)
(23)		5%	DB	89.7	90.4	91.4	91.2	91.3	90.4	89.9	91.1	91.6	92.2	91.1	90.5	(23)
(24)			MCWB	76.9	76.9	76.8	77.4	78.0	77.3	76.9	76.7	76.7	77.1	77.4	77.1	(24)
(25)		10%	DB	88.1	88.5	89.6	89.5	89.6	88.5	88.2	89.1	89.8	90.2	89.4	88.7	(25)
(26)			MCWB	76.8	76.7	76.8	77.2	77.7	77.2	76.8	76.5	76.7	77.1	77.2	76.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.7	79.5	79.6	79.8	80.5	80.1	79.8	79.6	80.0	79.9	79.7	79.6	(27)
(28)			MCDWB	89.1	89.3	90.1	89.3	90.4	89.0	89.2	89.4	89.9	90.4	90.0	89.5	(28)
(29)		2%	WB	78.6	78.4	78.7	79.0	79.7	79.2	78.8	78.6	78.9	79.0	78.8	78.7	(29)
(30)			MCDWB	87.7	87.7	88.6	88.2	88.9	87.5	87.9	88.2	88.6	89.1	88.6	88.3	(30)
(31)		5%	WB	78.0	77.8	78.1	78.4	79.1	78.5	78.1	77.9	78.1	78.3	78.2	78.1	(31)
(32)			MCDWB	86.5	86.7	87.3	87.1	87.7	86.4	86.6	87.0	87.4	87.9	87.3	87.0	(32)
(33)		10%	WB	77.3	77.2	77.4	77.9	78.5	77.9	77.4	77.3	77.4	77.8	77.7	77.4	(33)
(34)			MCDWB	85.3	85.6	86.1	86.2	86.5	85.4	85.3	85.8	86.3	86.9	86.2	85.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.4	14.3	14.8	14.6	14.7	14.1	14.4	15.3	15.7	15.5	14.5	13.9	(35)
(36)			MCDBR	16.0	16.8	17.2	16.5	16.8	16.9	16.8	17.6	17.9	17.8	16.5	16.2	(36)
(37)			MCWBR	5.3	5.4	5.4	4.8	5.4	5.3	5.5	5.6	5.7	5.6	4.8	5.1	(37)
(38)		5% WB	MCDWB	14.2	15.0	15.2	14.8	14.9	14.2	14.7	15.3	15.8	15.7	14.9	14.5	(38)
(39)			MCWBR	5.4	5.5	5.4	5.0	5.3	5.3	5.5	5.6	5.9	5.5	5.0	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.418	0.425	0.425	0.418	0.418	0.416	0.424	0.459	0.491	0.554	0.474	0.428	(40)	
(41)		taud	2.449	2.422	2.427	2.440	2.456	2.483	2.458	2.341	2.240	2.038	2.279	2.420	(41)	
(42)		Ebn at Noon	290	289	286	281	273	270	269	266	264	250	271	285	(42)	
(43)		Edh at Noon	37	39	39	37	35	33	35	40	46	56	44	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1530	1628	1673	1661	1586	1487	1548	1663	1789	1713	1629	1520	(44)	
(45)		RadStd	91	117	107	79	65	94	129	110	100	105	74	109	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.72	+0.47	+1.39	N/A	+0.76	+0.67	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		+0.72	+0.47	+1.39	N/A	+0.76	+0.67	N/A	N/A	N/A	N/A	

Nomenclature: See separate page